8	**(i)**		$\frac{dx}{dt} = \frac{(1+t^3)-t(3t^2)}{(1+t^3)^2}$	**M1**	**1.1a**	Use of quotient rule – may be gained for $\frac{dy}{dt}$ if $\frac{dx}{dt}$ not seen (allow ±)	
			$\frac{dx}{dt} = \frac{(1-2t^3)}{(1+t^3)^2}$ oe	**A1**	**1.1**		
			$\frac{dy}{dt} = \frac{2t(1+t^3)-t^2(3t^2)}{(1+t^3)^2}$				
			$\frac{dy}{dt} = \frac{(2t-t^4)}{(1+t^3)^2}$ oe	**A1**	**1.1**		
			$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dt}{dx} = \frac{2t-t^4}{1-2t^3}$	**M1**	**1.1a**	Substitution into *their* $\frac{dy}{dx}$ dep on earlier M1	
			$t=1 \Rightarrow \frac{dy}{dx} = -1$	**A1**	**2.1**		
				[5]			
8	**(ii)**		LHS = $\frac{t^3+t^6}{(1+t^3)^3}$ oe	**M1**	**1.1a**	**AG** Expression for LHS	
			$= \frac{t^3(1+t^3)}{(1+t^3)^3}$	**M1**	**1.1**	Factorising	
			RHS = $\frac{t^3}{(1+t^3)^2}$ = LHS	**A1**	**2.1**	Completion of argument	
				[3]			